

EECE4632: Spring 2026

Hardware-Software Codesign for FPGA-Based Systems

Class Time: Mon & Wed 2:50 – 4:30 pm

Class Room: 9C Hayden Labs

Instructor: Prof. Miriam Leeser

Email: m.leeser@northeastern.edu

Office Hours: Mon & Wed 1:45 – 2:45 pm in 9C Hayden

Teaching Assistant: Yicheng Qian Email: gian.yich@northeastern.edu

TA Office hours: Mon & Wed 4:30 – 5:30 in 9C Hayden

Course Overview:

EECE 4632 is a project-based course that revolves around the Xilinx Pynq platform <http://www.pynq.io/>. Students will implement HW/SW designs that involve using Python to write embedded processor code, and using FPGA fabric to implement the hardware designs. Students will use C/C++ to describe their hardware designs, and Python to control them in software. Topics covered will include HW/SW codesign, buses and interfacing, and high-level synthesis for hardware. Students will develop a series of designs in class, culminating in a project of their choosing. Potential project topics include (but are not limited to): audio processing, computer vision, cryptography, image processing, robotics, and wireless communications.

Concepts covered will include Hardware/Software co-design, processor interfacing, memory hierarchies and advanced architecture topics including pipelining. Designs will be implemented with a combination of software running on an embedded ARM processor and Xilinx Field Programmable Gate Array hardware.

Prerequisites

EECE 2322: Fundamentals of Digital Design and Computer Organization. Familiarity with Python and Jupyter Notebooks is helpful but not required.

Textbook

There is no textbook for this class. Online materials will be shared with students during the semester. See Leganto Course materials on Canvas.

THIS IS AN IN PERSON, HANDS ON Lab course. In person attendance is required for all class meetings.

Software for EECE 4632

This course will use version 2024.1 of the Xilinx Vitis tool suite. Please do not use a different version of the tools. Software for the class is available on the PCs in 9C Hayden. It may also be available on the COE VLAB. More information on using the VLAB is provided in a separate handout. Software for PYNQ is available on an SD card that will be provided in class.

Class Communications:

If you have a question of general interest to the class, please post it to the discussion boards on Canvas. Discussions are not graded and are there for you to get help with questions. You are encouraged to answer questions posted by others. If you email me a question that I think will be of benefit to the entire class, I will ask you to post your question to the discussion boards.

If you have a question about your individual grade or situation, feel free to email me at m.leeser@northeastern.edu. I try to respond to emails promptly, but it may be a day or two before you receive a response. ALWAYS PUT EECE4632 in the subject of your email.

I generally do not answer questions between 9:00 pm and 8:30am.

Grading

Your final grade is calculated as a numeric grade between 0 and 100 based on the percentages shown below:

Weights

- | | |
|-----------------------------------|------|
| • Labs, Lab Quizzes, and Homework | 30 % |
| • Quizzes | 30 % |
| • Final project | 40 % |

Class Organization

This course is designed to be an integrated lab and lecture class. Hardware will be available in 9 Hayden. [Keep your SD card with your board! The software that you develop will be on this card.](#) We have programmed it with the PYNQ image you will be using.

Several homework and lab experiments will be assigned. You will be asked to demonstrate your work to the instructor or TA in person. Short quizzes for the lab assignments may also be added.

Students are expected to complete a project of their choice with the agreement of the instructor. At the end of the semester, students are expected to demonstrate their projects, write a project report and do a project presentation. Groups of two for projects are encouraged. The role of each student needs to be clearly defined and agreed to with the instructor. A plan regarding how you will collaborate and share updates is also required.

All work in this course, including homework, lab assignments, and quizzes is expected to represent individual effort. Any violations of the Northeastern University code of conduct (as described in the Student Handbook) will be referred to the Office of Student Conduct & Conflict Resolution. Late homework will not be accepted unless a student requests an extension *before* the due date and it is approved by the instructor.

The issuance of incomplete grades will strictly follow the College of Engineering guidelines. An incomplete will only be given for missed work at the end of the term due to illness.

Quizzes

There will be a total of 3 quizzes during the semester. Quizzes will have an approximate duration of 30 minutes. There are no make-up quizzes unless a student has exceptional circumstances. Quizzes will **be in person**.

Topics Covered

Embedded systems design. Hardware/software codesign. FPGA architecture. Memory hierarchy. Processor interfacing. Pipelining. High-level synthesis. The PYNQ design flow. Linux and Python programming. Computer arithmetic.

Course Policies

Recording of Classes

Classes will only be recorded if a student contacts the instructor in advance with a legitimate excuse for not attending (such as illness).

Attendance

Attendance is an important element for success in class. It is required, unless you are unable to make it due to illness or other urgent or emergent reasons. Contact me via email *before* the class time for allowed absence, unless you are unable to. If you anticipate an extended absence due to the need to quarantine, we will need to figure out how you can make up the lab experiments that are an integral part of this class.

Use of AI in EECE 4632

By default, students are not permitted to use AI tools. Individual assignments where AI use is allowed will explicitly state this fact. Whenever AI tools (such as Claude, ChatGPT, Copilot, or similar software) are used in this course, their use must be clearly acknowledged by the student. Whenever you use an AI tool, you must explain which tool you used, what you used it for, and how it contributed to your work. Failure to disclose AI assistance will be considered a violation of academic integrity. All submitted work must reflect your own understanding, analysis, and learning.

Academic Integrity

A commitment to the principles of academic integrity is essential to the mission of Northeastern University. The promotion of independent and original scholarship ensures that students derive the most from their educational experience and their pursuit of knowledge. Academic dishonesty violates the most fundamental values of an intellectual community and undermines the achievements of the entire University.

As members of the academic community, students must become familiar with their rights and responsibilities. In each course, they are responsible for knowing the requirements and restrictions regarding research and writing, examinations of whatever kind, collaborative work, the use of study aids, the appropriateness of assistance, and other issues. Students are responsible for learning the conventions of documentation and acknowledgment of sources in their fields. Northeastern University expects students to complete all examinations, tests, papers, creative projects, and assignments of any kind according to the highest ethical standards, as set forth either explicitly or implicitly in this Code or by the direction of instructors.

Go to <https://catalog.northeastern.edu/handbook/policies-regulations/academic-integrity/> to access

the full academic integrity policy.

Student Accommodations

Northeastern University and Disability Access Services (DAS) are committed to providing disability services that enable students who qualify under Section 504 of the REHABILITATION ACT and THE AMERICANS WITH DISABILITIES ACT AMENDMENTS ACT (ADAAA) to participate fully in the activities of the university. To receive accommodations through DAS, students must provide documentation of a disability that demonstrates a current substantial limitation. Accommodations are approved based on a review of the information that is submitted and reviews are done on a case-by-case basis.

For more information, visit <https://disabilityaccessservices.sites.northeastern.edu/>.

Outreach, Engagement, Belonging

Northeastern University is committed to fostering a community of belonging, which is essential to the advancement of Northeastern University's mission of teaching and research. Our university is stronger as a result of the varied backgrounds, experiences, and perspectives that all members of our global community bring to the pursuit of knowledge. Embracing this pluralism is not the work of one office, department, or academic unit. It is a shared responsibility that spans disciplines and boundaries. By harnessing the power of our differences, we will continue to light the path to bold new ideas and life-changing discoveries.

It is my intention that students from all backgrounds and perspectives will be well served by this course, and that the diverse experiences that students bring to this class will be viewed as an asset. I welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, socioeconomic background, family education level, ability – and other visible and nonvisible differences. All members of this class are expected to contribute to a respectful, welcoming and belonging environment for every other member of the class. Your suggestions are encouraged and appreciated.

Please visit [Belonging at Northeastern](#) for more information.

Policy on Sexual and Gender-Based Harassment and Title IX

The Northeastern University [Policy on Sexual and Gender-Based Harassment and Title IX](#) articulates how the University will respond to reported allegations of Sexual Harassment, Quid Pro Quo Harassment, Sexual Assault, Domestic Violence, Intimate Partner Violence, Dating Violence, Stalking, Gender-Based Harassment and Retaliation, and provides a consolidated statement of the rights and responsibilities under University policies and Title IX, as amended by the Violence Against Women Reauthorization Act of 2013 and the U.S. Department of Education and Office for Civil Rights' revised regulations in 2020 to the Title IX of the Education Amendments of 1972 (Title IX), 20 U.S.C. § 1681. The policy describes how areas within the University will coordinate the provision of interim remedies and the prompt and effective investigation of allegations of Prohibited Offenses. This policy applies to all members of the University community, including students and prospective students, employees and prospective employees, faculty, staff and volunteers in connection with university activities. It further applies to on- or off-campus behavior involving students. Please visit <http://www.northeastern.edu/ouec> for a complete list of reporting options and further information.